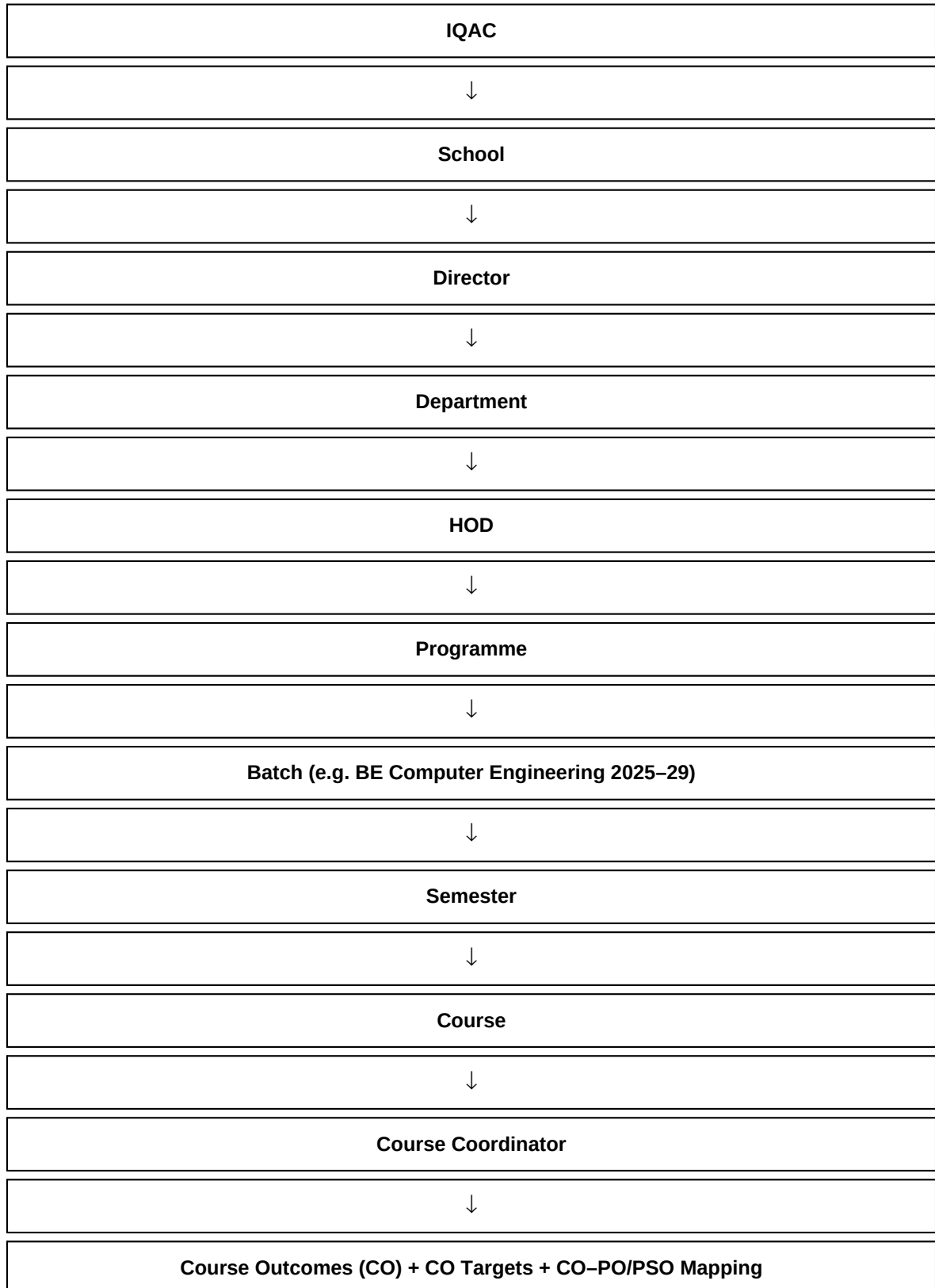


# New System Workflow

## DYPIU NBA Attainment System — Academic Year + Batch-Centred Architecture

**Core principle:** Academic Year is the main user-facing selector, while Batch is the central historical and calculation scope maintained by the system.

### 1. Overall Hierarchy



## 2. Role-Based Operational Workflow

### IQAC

- Has access to all Schools and their academic records.
- Views consolidated academic, attainment, ATR and approval records.
- Acts as the top-level institutional oversight layer.

### Director

- Works within the assigned School.
- Creates Departments under the School.
- Assigns the HOD for each Department.
- Creates/oversees Programmes under the Department.

### HOD

- Manages the Department's Programmes.
- Assigns Programme Coordinators to the relevant Programmes.
- Initializes and manages Batches, e.g. BE Computer Engineering 2025–29.
- Creates and maintains PO, PSO and PEO for the Programme.
- Creates/manages Courses for the applicable Batch/Semester.
- Reviews and approves Programme ATR.

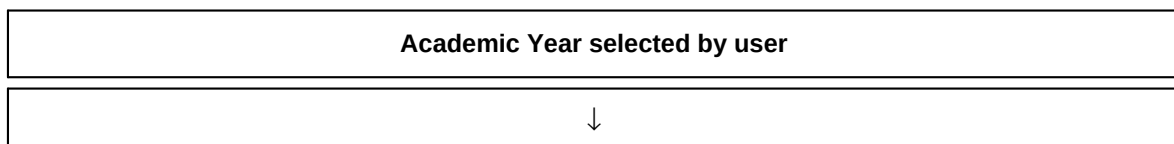
### Programme Coordinator

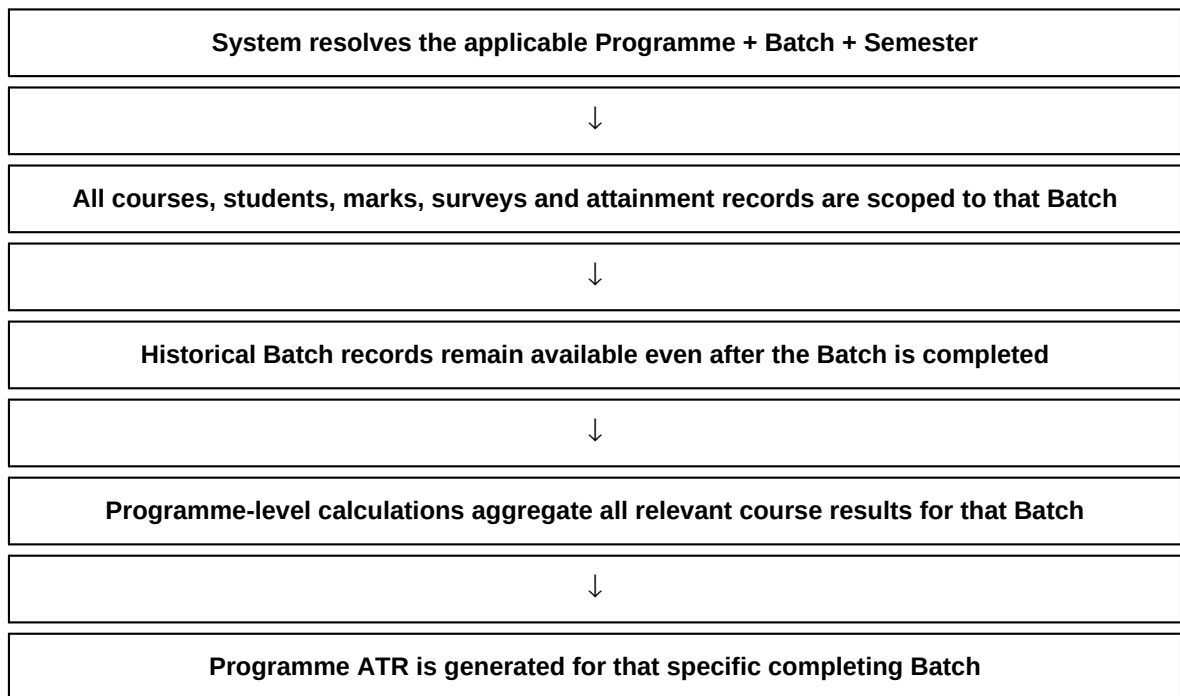
- Works only within the assigned Programme.
- Manages Programme-level setup.
- Sets PO/PSO targets.
- Receives/uses course-level attainment results across the Programme.
- Creates the final Programme ATR for the completing Batch.
- Submits Programme ATR to HOD for verification.

### Course Coordinator

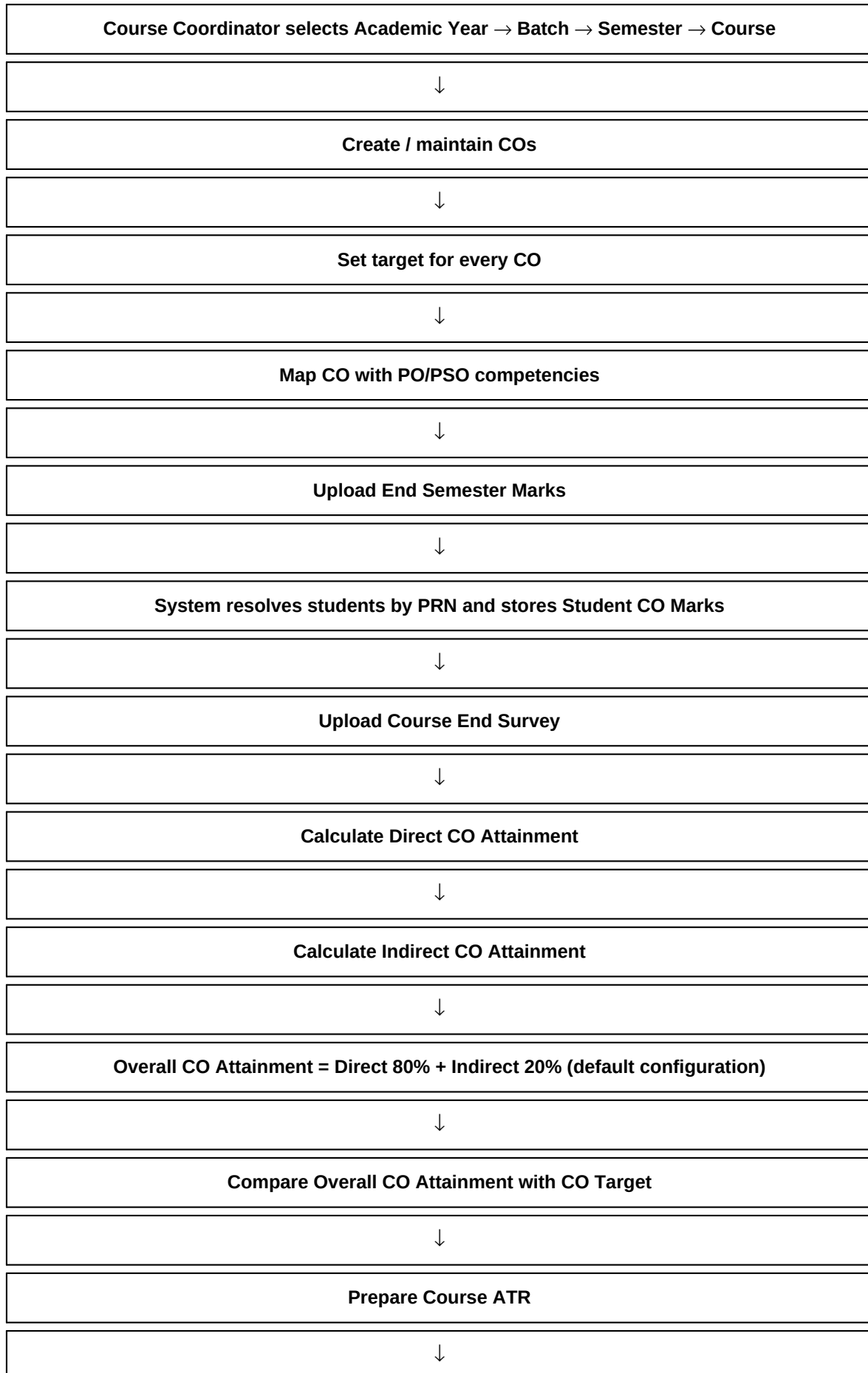
- Works within the assigned Course and Batch/Semester scope.
- Defines Course Outcomes (CO).
- Sets target for each CO.
- Creates CO–PO/PSO competency mappings.
- Uploads End Semester Marks.
- Uploads Course End Survey.
- Runs/initiates CO attainment calculation.
- Completes Course ATR using actual attainment versus CO target.

## 3. Batch-Centred Data Workflow



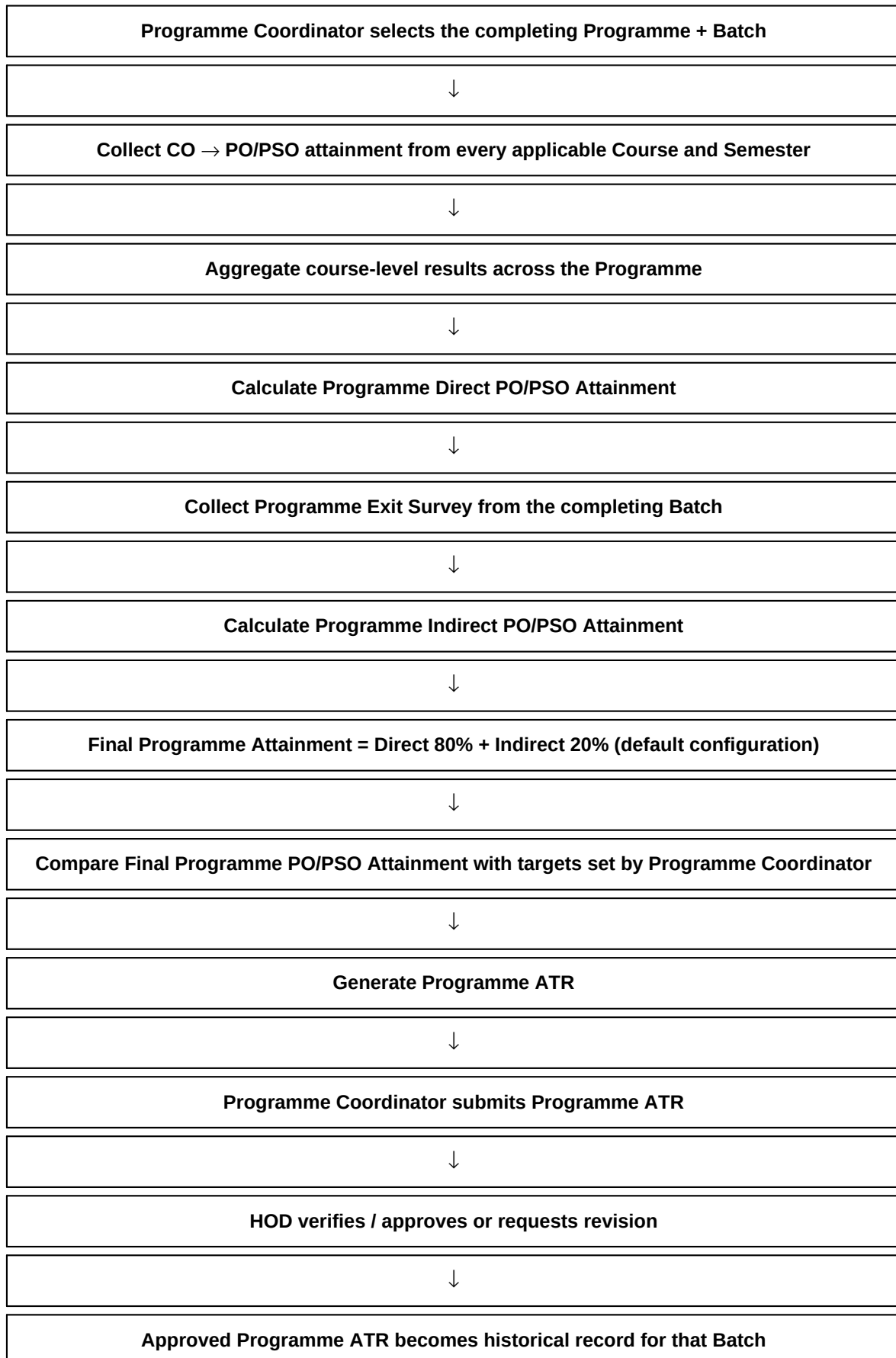


## 4. Course-Level Attainment Workflow



Submit / review / approve according to configured approval workflow

## 5. Programme Attainment Workflow



## 6. Attainment Data Lineage

| Level              | Source                                 | Calculation / Use                                                         |
|--------------------|----------------------------------------|---------------------------------------------------------------------------|
| Direct CO          | End Semester Marks → Student CO Marks  | Students meeting configured CO threshold → Direct CO % → Attainment Level |
| Indirect CO        | Course End Survey                      | Survey response weights → Indirect CO % → Attainment Level                |
| Overall CO         | Direct CO + Indirect CO                | Direct Level × 80% + Indirect Level × 20%                                 |
| PO/PSO Competency  | Overall CO + CO–PO/PSO Mapping         | Average/aggregation of mapped Overall CO results                          |
| PO/PSO             | PO/PSO Competencies                    | Aggregation of competency attainment                                      |
| Programme Direct   | Course-level PO/PSO results            | Aggregate all applicable courses/semesters for the Batch                  |
| Programme Indirect | Programme Exit Survey                  | Indirect Programme PO/PSO evidence                                        |
| Final Programme    | Direct + Indirect Programme Attainment | Direct 80% + Indirect 20%                                                 |
| Programme ATR      | Final Programme Attainment + Target    | Compare result with target and record observation/action plan             |

## 7. Approval Workflow

Course ATR: Course Coordinator → Submission → configured reviewer → Approved / Revision Required.

Programme ATR: Programme Coordinator → HOD → Approved / Revision Required. Every status transition is stored in Approval History.

Approved ATRs are treated as historical snapshots and should not be silently overwritten by later calculations.

## 8. Core Design Rules

- Academic Year is the primary UI filter.
- Batch is the central historical scope.
- Every assessment/attainment record must retain Batch context.
- Completed Batches remain queryable.
- Calculation Runs preserve historical results.
- Programme ATR is created only when required programme-level attainment and target data are available.